

- SOC Analyst Home Lab -

This guide will create 3 virtual machines (VMs) that will work together to form a SOC analyst training environment that can be used at home to help get some hands-on experience with working in a SOC environment. The 3 VMs that will be created are a Splunk Server, Windows 11 Endpoint, and a Domain Controller. The idea is to simulate a real domain environment that will create various simulated events such as successful user logins, failed logins, account lockouts, brute force attempts, suspicious PowerShell commands being executed, and more. These events will be generated automatically and can then be analysed via Splunk, using a combination of dashboards, triggered alerts that will be sent to a slack channel, and indexing.

Text in this document is colour coded:

Blue: Code/PowerShell cmdlets

Orange: Text that may change for your setup e.g. usernames/passwords

Green: Website address

Pink: File/file location

Red: Splunk search

All resources to create this lab can be found here:

<https://github.com/Jkelly515/soc-analyst-home-lab>

Contents

Introduction.....	1
Host Machine Hardware Requirements	1
VM Overview.....	1
1 - VirtualBox Install/Setup	2
1.1 - Install VirtualBox.....	2
1.2 - Create the Host-Only Network.....	2
1.3 - Set Up a VMShare Shared Folder	2
1.4 - Disable DHCP Server	2
2 - Ubuntu 24.04 LTS VM Install/Setup	3
2.1 - Download Ubuntu Server 24.04 LTS	3
2.2 - Create the Ubuntu VM in VirtualBox	3
2.3 - Configure the Ubuntu VM Settings	3
2.4 - Set the Static IP on the Host-Only Adapter	3
2.5 - Install SSH and Update	4
2.6 - Create Snapshot	4
3 - Splunk Enterprise on Ubuntu	5
3.1 - Create a Free Splunk Account	5
3.2 - Download Splunk Enterprise	5
3.3 - Install and Start Splunk	5
3.4 - Configure the Firewall	5
3.5 - Access Splunk Web.....	6
3.6 - Configure Splunk to Receive Data.....	6
3.7 - Create Indexes.....	6
3.8 - Correct Time Zone.....	6
4 – Windows 11 Pro VM	6
4.1 - Download Windows 11 ISO	7
4.2 - Create the Windows 11 VM.....	7
4.3 - Install Windows 11	8
4.4 - Windows 11 Initial Setup	8
4.5 - Verify Windows 11 is Pro Edition	8
4.6 - Enable copy and paste & drag and drop to and from the VM.....	9
4.7 - Set Static IP via Network Settings.....	9
4.8 - Enable SSH via Optional Features (Recommended but not essential)	9

4.9 - Map the Shared Folder	10
5 - Windows 11 Logging Configuration	10
5.1 - Enable Advanced Audit Policies.....	10
5.2 - Enable Command-Line Process Auditing	10
5.3 - Increase Event Log Sizes via Event Viewer	11
5.4 - Enable PowerShell Logging.....	11
6 - Sysmon Installation.....	11
6.1 - Download Sysmon and Config.....	11
6.2 - Install Sysmon	12
7 - Create Lab Users.....	12
7.1 - Run the User Creation Script	12
8 - Splunk Universal Forwarder on Windows 11	14
8.1 - Download and Transfer the Universal Forwarder	14
8.2 - Install via GUI.....	14
8.3 - Configure inputs.conf	15
8.4 - Configure outputs.conf	16
8.5 - Run the Forwarder as Local System via Services	16
8.6 - Verify Data is Flowing	16
9 - Splunk Add-ons.....	17
9.1 - Download Add-ons from Splunkbase.....	17
9.2 - Upload Each Add-on	17
10 - Windows Server 2022 VM (Domain Controller)	17
10.1 - Download Windows Server 2022 Evaluation.....	17
10.2 - Create the Windows Server VM.....	17
10.3 - Install Windows Server 2022.....	18
10.4 - Configure Network via Network Settings	18
10.5 - Update the Time Zone (if needed)	19
10.6 - Rename the Server via System Properties	19
10.7 - Install Guest Additions and Map the Shared Folder	19
10.8 - Allow Communication between the VMs.....	19
11 - Active Directory Domain Services	20
11.1 - Install AD DS and DNS via Server Manager	20
11.2 - Promote to Domain Controller via the Wizard.....	20
11.3 - Verify Active Directory	20

11.4 - Configure Account Lockout Policy via Group Policy.....	21
12 - Create Domain Users and Organisational Units.....	21
12.1 - Create OU Structure via Active Directory Users and Computers	21
12.2 - Transfer lab-users.csv to the DC.....	21
12.3 - Create Domain Users from the CSV	22
13 - Join Windows 11 to the Domain	23
13.1 - Configure DNS to Point to the Domain Controller	23
13.2 - Join the Domain via System Properties.....	23
13.3 - Verify Domain Membership	24
13.4 - Restore NAT Adapter DNS	24
13.5 - Install RSAT Active Directory Tools.....	24
14 - Splunk Universal Forwarder on DC01	24
14.1 - Install the Splunk UF onto the DC	24
14.2 - Configure inputs.conf via Notepad	25
14.3 - Configure outputs.conf via Notepad	26
14.4 - Run Forwarder as Local System via Services	26
14.5 - Verify DC Logs are Flowing.....	26
15 - Domain Activity Simulator.....	27
15.1 - Copy the Domain Simulator to Windows 11	27
15.2 - Test the Script.....	27
15.3 - Register the Domain Simulator Scheduled Task	27
16 – Splunk Dashboards and Alerts.....	28
16.1 - Important Note About Domain Login Events.....	28
16.2 - Create Dashboard.....	28
16.3 - Create Alerts.....	29
17 - Slack Workplace for Splunk Alerts.....	30
17.1 - Create Channel.....	30
17.2 - Create Webhook	30
17.3: Create the slack_alert.sh script	30
Conclusion	32

Introduction

I started this project to give myself a starting point when it comes to working in a SOC environment. I noticed when applying for jobs (and through my own life experience) that having a solid understanding of concepts and theory is valuable and certainly a great place to start, but having actual hands-on experience with those concepts and seeing how they work in practice is what turns that knowledge into a demonstratable skill. Many employers want to hire someone with experience, however for many that becomes a catch-22, because they need the job to get experience but can't get the job because they don't have experience. This project is my attempt to break that paradox.

Host Machine Hardware Requirements

Resource	Minimum	Recommended
RAM	24 GB	32 GB
CPU	6 cores with VT-x/AMD-V enabled	8+ cores
Free Storage	200GB SSD	300GB SSD
Host OS	Windows 10 or Windows 11	Windows 11

Note: All 3 VMs will be created with a disk size of 80GB. If storage space is tight, this can be reduced to 40GB for the Splunk Server, and 60GB each for the Windows Endpoint and Domain Controller.

VM Overview

VM	OS	Role	IP
Splunk Server	Ubuntu 24.04 LTS	Splunk Enterprise (SIEM/indexer)	192.168.56.10
Windows Endpoint	Windows 11 Pro	Monitored workstation + Splunk UF	192.168.56.11
Domain Controller (DC)	Windows Server 2022	Active Directory (TestDomain.ie)	192.168.56.12

1 - VirtualBox Install/Setup

1.1 - Install VirtualBox

1. Go to [virtualbox.org/wiki/Downloads](https://www.virtualbox.org/wiki/Downloads)
2. Click Windows hosts under 'VirtualBox Platform Packages' and allow it to download
3. Click Accept and download under 'VirtualBox Extension Pack'
4. Run the VirtualBox installer, accept all defaults. Click Yes when warned about a brief network disconnect (this is normal)
5. After VirtualBox installs, double-click the Extension Pack .vbox-extpack file and accept the licence

1.2 - Create the Host-Only Network

VirtualBox 7.2 removed the Network Manager from the GUI. You must use PowerShell on your host PC instead. First, check if a host-only adapter already exists:

1. Press Win + R, type `ncpa.cpl`, press Enter
2. Look for an adapter called VirtualBox Host-Only Ethernet Adapter
3. If it exists, open PowerShell as Administrator on your host PC and verify its IP:

`& "C:\Program Files\Oracle\VirtualBox\VBXManage.exe" list hostonlyifs`

If you see IPAddress: 192.168.56.1 you can skip to Step 1.3

If not, create it by running these commands in order:

`& "C:\Program Files\Oracle\VirtualBox\VBXManage.exe" hostonlyif create`

`& "C:\Program Files\Oracle\VirtualBox\VBXManage.exe" hostonlyif ipconfig "VirtualBox Host-Only Ethernet Adapter" --ip 192.168.56.1 --netmask 255.255.255.0`

`& "C:\Program Files\Oracle\VirtualBox\VBXManage.exe" dhcpserver add --ifname "VirtualBox Host-Only Ethernet Adapter" --ip 192.168.56.1 --netmask 255.255.255.0 --lowerip 192.168.56.100 --upperip 192.168.56.200 --enable`

1.3 - Set Up a VMShare Shared Folder

Drag and drop between two guest VMs is a known bug in VirtualBox 7.2 and does not work reliably. A shared folder on your host PC is the best way to transfer files between VMs.

To do this, create a new folder on your host PC at C: - name it 'VMShare'. You will add this to each VM after creating them to allow for easy file transfer.

1.4 - Disable DHCP Server

In VirtualBox, select the Network tab on the left. Near the bottom there is a tab called DHCP Server, ensure the Enable Server checkbox is unticked

2 - Ubuntu 24.04 LTS VM Install/Setup

2.1 - Download Ubuntu Server 24.04 LTS

Go to ubuntu.com/download/server and download Ubuntu Server 24.04 LTS, under the 'Previous releases' section. It should be a ~3.2 GB ISO.

2.2 - Create the Ubuntu VM in VirtualBox

1. Open VirtualBox and click New, fill in these details (leave others as default):
 - VM Name: SplunkServer
 - ISO Image: browse to your Ubuntu .iso file
 - Type: Linux
 - Version: Ubuntu (64-bit)
 - Leave 'Proceed with Unattended Installation' ticked
2. Set up unattended guest OS installation:
 - Create whatever username and password you like, leave OS Installation Options as default.

*Note: For this guide, I have chosen the username **UbuntuAdmin**. If you have chosen a different username, replace any instances of **UbuntuAdmin** with the username you have chosen instead*
3. Specify virtual hardware:
 - Base Memory 4096 MB, Processors 2
4. Specify virtual hard disk:
 - Hard Disk 80 GB
5. Finish - If the VM automatically boots up, power it off

2.3 - Configure the Ubuntu VM Settings

1. Right-click the SplunkServer VM, Settings, Network:
 - Adapter 1: NAT (leave as default)
 - Adapter 2: tick Enable Network Adapter, Attached to Host-only Adapter, select VirtualBox Host-Only Ethernet Adapter
2. Settings, Shared Folders, click the + icon:
 - Folder Path: C:\VMShare
 - Folder Name: VMShare
 - Tick Auto-mount
3. Start the SplunkServer VM and wait for the install to complete

2.4 - Set the Static IP on the Host-Only Adapter

NOTE: You will need to manually type out each command until after step 2.6 is complete.

1. Log in at the console, then find your netplan config:

`ls /etc/netplan/`

In my instance, mine was '50-cloud-init.yaml', yours may be different.

2. Edit the file shown:

```
sudo nano /etc/netplan/50-cloud-init.yaml
```

Update it to include both adapters (keep whatever exists for enp0s3, add the enp0s8 section):

```
network:
```

```
  version: 2
```

```
  ethernets:
```

```
    enp0s3:
```

```
      dhcp4: true
```

```
    enp0s8:
```

```
      dhcp4: no
```

```
      addresses: [192.168.56.10/24]
```

NOTE: YAML requires consistent indentation with spaces, never tabs. Each level must align exactly.

Press Ctrl+X, then Y, then enter.

3. Apply:

```
sudo netplan apply
```

```
ip addr show enp0s8
```

You should see 192.168.56.10 on enp0s8.

2.5 - Install SSH and Update

```
sudo apt install -y openssh-server
```

```
sudo systemctl enable ssh --now
```

```
sudo ufw allow 22/tcp
```

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt install -y curl wget net-tools ufw
```

You can now SSH in from your host machine which will allow you to copy+paste commands from this guide going forward, just ensure that the SplunkServer VM is still running in the background. To do this, open a PowerShell terminal on your host machine and enter in this command:

```
ssh UbuntuAdmin@192.168.56.10
```

2.6 - Create Snapshot

In the VirtualBox Menu for SplunkServer, click Machine, Take Snapshot..., name it "Clean Ubuntu Install"

3 - Splunk Enterprise on Ubuntu

3.1 - Create a Free Splunk Account

Go to splunk.com, Free Splunk, Create account. You need this to download Splunk and add-ons.

Splunk Enterprise is free for home labs. The free tier allows 500 MB of data per day. After any trial period it reverts to the free tier and continues working

3.2 - Download Splunk Enterprise

Log into splunk.com, Trials & Downloads, Start Free Trial (under Splunk Enterprise), Linux. Beside .deb, click 'Copy wget link' and run it on SplunkServer with this command:

```
wget -O splunk-enterprise.deb "wget link that you just copied"
```

3.3 - Install and Start Splunk

```
sudo dpkg -i splunk-enterprise.deb  
sudo /opt/splunk/bin/splunk start --accept-license --run-as-root
```

When prompted, create an admin username and choose a strong password. For this guide, I have chosen the admin username '[splunkadmin](#)'. **If you have chosen a different username, replace any instances of '[splunkadmin](#)' in this guide with your chosen username**

Then fix ownership and configure autostart:

```
sudo chown -R UbuntuAdmin:UbuntuAdmin /opt/splunk  
sudo /opt/splunk/bin/splunk enable boot-start -user UbuntuAdmin
```

Note: Always use `sudo -u UbuntuAdmin` when running Splunk commands. Plain `sudo` causes ownership issues. For example, use:

```
sudo -u UbuntuAdmin /opt/splunk/bin/splunk start  
sudo -u UbuntuAdmin /opt/splunk/bin/splunk stop
```

Instead of:

```
sudo /opt/splunk/bin/splunk start  
sudo /opt/splunk/bin/splunk stop
```

3.4 - Configure the Firewall

```
sudo ufw allow 22/tcp comment "SSH"  
sudo ufw allow 8000/tcp comment "Splunk Web"  
sudo ufw allow 9997/tcp comment "Splunk Receiving"  
sudo ufw allow 8089/tcp comment "Splunk Management"  
sudo ufw enable  
sudo ufw status verbose
```

3.5 - Access Splunk Web

On your host machine browser go to:

<http://192.168.56.10:8000>

Log in with **splunkadmin** and your password

3.6 - Configure Splunk to Receive Data

Settings, Forwarding and Receiving, Configure Receiving, New Receiving Port

Enter 9997, Save

3.7 - Create Indexes

1. Settings, Indexes, New Index. Create each of these (leave all other settings at default):

Name

- windows-logs

- sysmon-logs

- dc-logs

3.8 - Correct Time Zone

1. Click the 'Administrator' dropdown menu at the top right
2. Select Preferences, then change the Time Zone setting to match your region's time zone
3. Take a VM snapshot: Splunk Enterprise Configured

3.9 - Commands to Take Note of

Sometimes Splunk Web may stall/stop working. Use these commands on the Ubuntu-SplunkServer VM to check the status of splunk, stop, and start it

`sudo -u UbuntuAdmin /opt/splunk/bin/splunk status`

`sudo -u UbuntuAdmin /opt/splunk/bin/splunk stop`

`sudo -u UbuntuAdmin /opt/splunk/bin/splunk start`

4 – Windows 11 Pro VM

4.1 - Download Windows 11 ISO

Go to microsoft.com/en-gb/software-download/windows11, Windows 11 Disk Image (ISO) for x64 devices, Download now, Select "English International" for the language, Confirm, 64-bit Download

The downloaded ISO should be ~7.9 GB

4.2 - Create the Windows 11 VM

1. Open VirtualBox and click New, fill in these details (leave others as default):

- VM Name: Windows11-Endpoint
- ISO Image: browse to your Ubuntu Windows 11 .iso file
- OS: Microsoft Windows
- OS Edition: Windows 11 Pro
- OS Version: Windows 11 (64-bit)
- Leave 'Proceed with Unattended Installation' ticked

2. Specify virtual hardware:

- Base Memory: 8192 MB
- Number of CPUs: 4
- Tick the Use EFI checkbox

3. Specify virtual hard disk:

- Disk Size: 80GB

4. Finish, right-click the Windows11-Endpoint VM, Settings:

System:

- TPM Version: 2.0

Network, Adapter 1:

- NAT

Network, Adapter 2:

- Enable Network Adapter
- Attached to: Host-only Adapter
- Name: VirtualBox Host-Only Ethernet Adapter

Shared Folders:

- Click the + icon (Add new shared folder)
- Folder Path C:\VMShare
- Folder Name VMShare
- tick Auto-mount

5. Click OK

4.3 - Install Windows 11

1. Start the Windows11-Endpoint VM, click inside the window and press any key immediately when you see "Press any key to boot from CD or DVD" (it only shows for a few seconds)
2. Select English (United Kingdom) for the Language, Time and Currency format, and Keyboard or input method (I am using English (United Kingdom))
3. Select the following in this order:
 - English (United Kingdom) for the Language, Time and Currency format
 - United Kingdom for the Keyboard or input method
 - Install Windows 11
 - Tick the "I agree to everything..." box
 - I don't have a product key
 - Windows 11 Pro
 - Accept the licence agreement
 - Disk 0 Unallocated Space
 - Install
4. Wait for the install to complete

4.4 - Windows 11 Initial Setup

1. Country: United Kingdom, Yes
2. Keyboard: United Kingdom, Yes, Skip second keyboard
3. Name your PC: WIN11-ENDPOINT, Next
4. Set up for personal use (the VM will likely update after this)
5. Press Shift + (fn)F10 and enter this command into the command prompt that opens:
[start ms-cxh:localonly](#)
6. Enter in a Username and Password. For this lab I will use the Username **WinAdmin**
7. Answer three security questions, then hit Next
8. Choose the following options in this order:
 - No
 - No
 - Required only
 - No
 - No

4.5 - Verify Windows 11 is Pro Edition

1. Start, Settings, System, About
2. Under Windows info, verify that the Edition is Windows 11 Pro

If it says Windows 11 Home, open PowerShell as Administrator and run:

[changepk.exe /productkey VK7JG-NPHTM-C97JM-9MPGT-3V66T](#)

Windows will then change version and restart. The key used here is a generic Pro key published by Microsoft and is legitimate for this purpose.

4.6 - Enable copy and paste & drag and drop to and from the VM

1. Using the VirtualBox menu bar on the Windows11-Endpoint VM, select Devices, Insert Guest Additions CD image
2. Open up **CD Drive (D):** in File Explorer and run the **VBoxWindowsAdditions-amd64** file, select Next, Next, Install.
3. Reboot the VM once install has completed
4. In the VirtualBox menu bar, select Devices, Shared Clipboard, Bidirectional. Also select Devices, Drag and Drop, Bidirectional

You should now be able to copy and paste & drag and drop to and from the VM and your host PC

*Note: Drag and drop between two guest VMs does not work in VirtualBox 7.2. For VM-to-VM file transfers always use the **VMShare** folder at **(Z:)***

4.7 - Set Static IP via Network Settings

1. Press Win + R (or search for "Run" in the search bar and click on the Run app), type ncpa.cpl, press Enter
2. You will see two adapters. Right-click Ethernet 2 (the host-only adapter), Properties
3. Select Internet Protocol Version 4 (TCP/IPv4), click Properties
4. Select Use the following IP address:
 - IP address: 192.168.56.11
 - Subnet mask: 255.255.255.0
 - Default gateway: leave blank
5. Click OK, Close

Test connectivity by opening PowerShell as Administrator:

ping 192.168.56.10

You should receive replies from the Splunk server. Ensure the SplunkServer VM is still running.

4.8 - Enable SSH via Optional Features (Recommended but not essential)

1. Start, Settings, System, Optional Features, View features, See available features
2. Search for and click OpenSSH Server, tick it and click Add (it is normal for it to stall and take quite a while to add)
3. After installation completes, open PowerShell as Administrator and run:

Start-Service sshd

Set-Service -Name sshd -StartupType Automatic

New-NetFirewallRule -Name sshd -DisplayName "OpenSSH Server" -Enabled True -Direction Inbound -Protocol TCP -Action Allow -LocalPort 22

From your host: **ssh WinAdmin@192.168.56.11**

After connecting with SSH, you land in CMD by default. Always type powershell and press Enter immediately to switch. All commands in this guide assume you are in PowerShell, not CMD

4.9 - Map the Shared Folder

Open PowerShell as Administrator and run:

```
net use V: \\vboxsvr\VMShare /persistent:yes
```

Files in **C:\VMShare** on your host appear at **V:** inside the VM.

Take a VM snapshot: Clean Windows 11 Install

5 - Windows 11 Logging Configuration

5.1 - Enable Advanced Audit Policies

Open PowerShell as Administrator and run:

```
auditpol /set /subcategory:"Credential Validation" /success:enable /failure:enable  
  
auditpol /set /subcategory:"Kerberos Authentication Service" /success:enable /failure:enable  
auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable  
  
auditpol /set /subcategory:"Security Group Management" /success:enable /failure:enable  
auditpol /set /subcategory:"Logon" /success:enable /failure:enable  
  
auditpol /set /subcategory:"Logoff" /success:enable /failure:enable  
  
auditpol /set /subcategory:"Account Lockout" /success:enable /failure:enable  
  
auditpol /set /subcategory:"Special Logon" /success:enable /failure:enable  
  
auditpol /set /subcategory:"File System" /success:enable /failure:enable  
  
auditpol /set /subcategory:"Registry" /success:enable /failure:enable  
auditpol /set /subcategory:"Audit Policy Change" /success:enable /failure:enable  
  
auditpol /set /subcategory:"Sensitive Privilege Use" /success:enable /failure:enable  
  
auditpol /set /subcategory:"Process Creation" /success:enable  
  
auditpol /set /subcategory:"Process Termination" /success:enable  
  
auditpol /set /subcategory:"Security System Extension" /success:enable /failure:enable  
  
Write-Host "Audit policies configured." -ForegroundColor Green
```

5.2 - Enable Command-Line Process Auditing

While still running PowerShell as Administrator, run:

```
$path = "HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\Audit"  
New-Item -Path $path -Force | Out-Null  
Set-ItemProperty -Path $path -Name "ProcessCreationIncludeCmdLine_Enabled" -Value 1 -Type  
DWORD -Force  
Write-Host "Command-line auditing enabled." -ForegroundColor Green
```

5.3 - Increase Event Log Sizes via Event Viewer

1. Press Win + R, type eventvwr.msc, press Enter
2. Expand Windows Logs in the left panel
3. Right-click Security, Properties, set Maximum log size to 1048576 (Which is 1 GB), OK
4. Right-click System, Properties: set to 262144 (256 MB), OK
5. Right-click Application, Properties: set to 262144 (256 MB), OK

5.4 - Enable PowerShell Logging

Open PowerShell as Administrator and run:

```
$sbPath = "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging"
New-Item -Path $sbPath -Force | Out-Null Set-ItemProperty -Path $sbPath -Name
"EnableScriptBlockLogging" -Value 1
```

```
$mlPath = "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ModuleLogging" New-
Item -Path $mlPath -Force | Out-Null Set-ItemProperty -Path $mlPath -Name
"EnableModuleLogging" -Value 1 New-Item -Path "$mlPath\ModuleNames" -Force | Out-Null Set-
ItemProperty -Path "$mlPath\ModuleNames" -Name "*" -Value "*"
```

```
$transPath = "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\Transcription" New-
Item -Path $transPath -Force | Out-Null Set-ItemProperty -Path $transPath -Name
"EnableTranscripting" -Value 1 Set-ItemProperty -Path $transPath -Name "OutputDirectory" -
Value "C:\PSTranscripts" Set-ItemProperty -Path $transPath -Name "EnableInvocationHeader" -
Value 1
```

```
New-Item -Path "C:\PSTranscripts" -ItemType Directory -Force | Out-Null New-Item -Path
"C:\Tools" -ItemType Directory -Force | Out-Null Write-Host "PowerShell logging enabled." -
ForegroundColor Green
```

6 - Sysmon Installation

6.1 - Download Sysmon and Config

Open PowerShell as Administrator and run:

```
Invoke-WebRequest -Uri "https://download.sysinternals.com/files/Sysmon.zip" -OutFile
"C:\Tools\Sysmon.zip" Expand-Archive -Path "C:\Tools\Sysmon.zip" -DestinationPath
"C:\Tools\Sysmon" -Force
```

```
Invoke-WebRequest -Uri "https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-
config/master/sysmonconfig-export.xml" -OutFile "C:\Tools\sysmonconfig.xml" Write-Host
"Sysmon downloaded." -ForegroundColor Green
```

6.2 - Install Sysmon

Open PowerShell as Administrator and run:

```
& "C:\Tools\Sysmon\Sysmon64.exe" -accepteula -i "C:\Tools\sysmonconfig.xml"
```

```
Get-Service Sysmon64
```

```
Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 5
```

Sysmon is designed to protect itself from being stopped. Never try Restart-Service Sysmon64 - it will always fail by design.

Take a VM snapshot: Windows 11 - Logging and Sysmon Configured

7 - Create Lab Users

7.1 - Run the User Creation Script

Open PowerShell as Administrator on Windows 11. Paste the entire script below and run:

```
$firstNames =  
@( "Jack","Oliver","Harry","George","Charlie","Jacob","Thomas","Oscar","William","James","Emily","  
Grace","Sophie","Amelia","Chloe","Lucy","Ellie","Charlotte","Megan","Katie","Liam","Ryan","Callum",  
"Connor","Cameron","Aaron","Finn","Dylan","Aidan","Fionn","Aoife","Niamh","Saoirse","Ciara","Orla",  
"Roisin","Sinead","Aisling","Maeve","Caoimhe","Daniel","Ethan","Joseph","Adam","Sean","Kieran","R  
hys","Owen","Bethan","Erin"  
  
)  
  
$lastNames =  
@( "Smith","Jones","Taylor","Williams","Brown","Davies","Evans","Wilson","Thomas","Roberts","John  
son","Lewis","Walker","Robinson","Wood","Thompson","White","Watson","Jackson","Wright","Green",  
"Harris","Cooper","King","Baker","Hughes","Clarke","Morgan","Hunt","Murphy","Kelly","Mulligan","Ry  
an","Walsh","Mullen","Byrne","McCarthy","Doyle","Fitzgerald","Gallagher",  
"MacDonald","Campbell","Stewart","Robertson","Fraser","Hamilton","Reid","Bell","Cunningham","Gr  
aham"  
  
)  
  
$departments =  
@("IT","HR","Finance","Marketing","Operations","Sales","Legal","R&D","Support","Management")  
  
function New-RandomPassword {  
    $upper = "ABCDEFGHJKLMNPQRSTUVWXYZ"  
    $lower = "abcdefghijklmnopqrstuvwxyz"  
    $digits = "23456789"  
    $special = "!@#$$%^&*"   
    $pass = $upper[(Get-Random -Maximum $upper.Length)]  
    $pass += $lower[(Get-Random -Maximum $lower.Length)]  
    $pass += $digits[(Get-Random -Maximum $digits.Length)]
```



```

$pass += $special[(Get-Random -Maximum $special.Length)]
$all = $upper + $lower + $digits + $special
for ($i = 0; $i -lt 8; $i++) { $pass += $all[(Get-Random -Maximum $all.Length)] }
return -join ($pass.ToCharArray() | Get-Random -Count $pass.Length)
}

$createdUsers = @()
Write-Host "Creating 30 lab users..." -ForegroundColor Cyan

for ($i = 1; $i -le 30; $i++) {
    $firstName = $firstNames | Get-Random
    $lastName = $lastNames | Get-Random
    $dept = $departments | Get-Random
    $base = ($firstName.Substring(0,1) + $lastName).ToLower()
    $username = $base
    $suffix = 1
    while (Get-LocalUser -Name $username -ErrorAction SilentlyContinue) {
        $username = "$base$suffix"; $suffix++
    }
    $password = New-RandomPassword
    $securePass = ConvertTo-SecureString $password -AsPlainText -Force
    try {
        New-LocalUser -Name $username -Password $securePass -FullName "$firstName $lastName" `
            -Description "SOC Lab - $dept" -PasswordNeverExpires -ErrorAction Stop
        Add-LocalGroupMember -Group "Users" -Member $username -ErrorAction SilentlyContinue
        $createdUsers += [PSCustomObject]@{
            Username = $username; FullName = "$firstName $lastName"
            Department = $dept; Password = $password; IsAdmin = "No"
        }
        Write-Host " [+] $username" -ForegroundColor Green
    } catch { Write-Host " [!] Failed: $username" -ForegroundColor Red }
}

$adminCandidates = $createdUsers | Get-Random -Count 3
foreach ($admin in $adminCandidates) {
    Add-LocalGroupMember -Group "Administrators" -Member $admin.Username -ErrorAction
    SilentlyContinue
    $admin.IsAdmin = "Yes"
    Write-Host " [*] Admin: $($admin.Username)" -ForegroundColor Yellow
}

$createdUsers | Export-Csv -Path "C:\Tools\lab-users.csv" -NoTypeInfoInformation
Write-Host "`nDone. Credentials at C:\Tools\lab-users.csv" -ForegroundColor Cyan

```

The password generator always includes uppercase, lowercase, numbers, and special characters to meet both local and domain policy requirements. You can see the list of generated users, their passwords, and whether or not they have an admin account with the [C:\Tools\lab-users.csv](#) file

8 - Splunk Universal Forwarder on Windows 11

8.1 - Download and Transfer the Universal Forwarder

1. On your host machine go to [splunk.com](#), Trials & Downloads, Universal Forwarder (Download). Download the 64-bit Windows 10, 11, Windows Server 2019, 2022, 2025 MSI
2. Copy the MSI to [C:\VMShare](#) on your host. Inside Windows 11 copy it from [V:\](#) to [C:\Tools](#).

8.2 - Install via GUI

Double-click the MSI file in [C:\Tools](#) and work through each screen:

Screen 1 - Licence:

- Tick Check this box to accept the License Agreement
- Leave An on-premises Splunk Enterprise instance selected
- Click Next

Screen 2 - Credentials:

- Username: [admin](#)
- Password: choose something, note it down (this is just for the forwarder's own management interface)
- Click Next

Screen 3 - Deployment Server:

- Leave completely blank
- Click Next

Screen 4 - Receiving Indexer:

- Hostname or IP: 192.168.56.10
- Port: 9997
- Click Add then Next

Screen 5 - Install, Finish

8.3 - Configure inputs.conf

1. Open Notepad as Administrator (search Notepad in Start, right-click, Run as administrator)
2. Open the file: `C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf` (If it does not exist, create it)
3. Paste in this content (overwrite any existing text):

[WinEventLog://Security]

index = windows-logs

disabled = 0

renderXml = false

[WinEventLog://System]

index = windows-logs

disabled = 0

renderXml = false

[WinEventLog://Application]

index = windows-logs

disabled = 0

renderXml = false

[WinEventLog://Microsoft-Windows-Sysmon/Operational]

index = sysmon-logs

disabled = 0

renderXml = false

[WinEventLog://Microsoft-Windows-PowerShell/Operational]

index = windows-logs

disabled = 0

renderXml = false

[WinEventLog://Microsoft-Windows-Windows Defender/Operational]

index = windows-logs

disabled = 0

renderXml = false

4. Save the file

8.4 - Configure outputs.conf

1. In the same folder, create or open outputs.conf and paste:

```
[tcpout]
```

```
defaultGroup = splunk_indexer
```

```
[tcpout:splunk_indexer]
```

```
server = 192.168.56.10:9997
```

```
compressed = true
```

```
[tcpout-server://192.168.56.10:9997]
```

2. Save the file.

8.5 - Run the Forwarder as Local System via Services

1. Press Win + R, type services.msc, press Enter
2. Scroll down to SplunkForwarder
3. Right-click, Properties
4. Click the Log On tab
5. Select Local System account
6. Click OK
7. Right-click SplunkForwarder, Stop
8. If it gives an error, open PowerShell as Administrator and run: taskkill /F /IM splunkd.exe
9. Right-click SplunkForwarder, Start

The Status column should show 'Running'.

8.6 - Verify Data is Flowing

In Splunk Web (on your host PC) at <http://192.168.56.10:8000>, go to Search and Reporting and run this search:

```
index=windows-logs earliest=-5m
```

If events appear within a couple of minutes, data is flowing correctly.

Take snapshots of both VMs: Full Lab - Forwarder Configured

9 - Splunk Add-ons

9.1 - Download Add-ons from Splunkbase

On your host machine go to splunkbase.splunk.com and log in with your Splunk account. Download each of these:

1. Splunk Add-on for Microsoft Windows - normalises Windows Event Log fields
2. Splunk Add-on for Sysmon - normalises Sysmon event fields
3. Splunk Security Essentials - pre-built detection use cases

9.2 - Upload Each Add-on

In Splunk Web:

1. Apps (the 9 dots icon at the top right), Manage
2. Install app from file
3. browse..., select one of the three files you just downloaded
4. Upload
5. Repeat for the other two add-ons. Splunk Security Essentials will download as a .tgz file, the entire file can be uploaded

After all three are uploaded, restart Splunk on the SplunkServer VM:

```
sudo -u UbuntuAdmin /opt/splunk/bin/splunk stop
```

```
sudo -u UbuntuAdmin /opt/splunk/bin/splunk start
```

10 - Windows Server 2022 VM (Domain Controller)

10.1 - Download Windows Server 2022 Evaluation

Go to info.microsoft.com/ww-landing-windows-server-2022.html and fill in the free registration form. For the Company Name you can use "Personal Lab", for size use "1" and for Job Role use "Student". Download the ISO (64-bit, approximately 4.7 GB).

10.2 - Create the Windows Server VM

1. Open VirtualBox and click New, fill in these details (leave others as default):
 - VM Name: DC01-WindowsServer
 - ISO Image: Select the ISO downloaded from the previous step
 - OS: Microsoft Windows
 - OS Version: Windows Server 2022 (64-bit)
 - Leave 'Proceed with Unattended Installation' ticked
2. Specify virtual hardware:
 - Base Memory: 4096 MB
 - Number of CPUs: 2

3. Specify virtual hard disk:
 - Disk Size: 80GB
4. Finish, Right-click the DC01-WindowsServer, Settings:

Network, Adapter 1:

- NAT

Network, Adapter 2:

- Enable Network Adapter
- Attached to: Host-only Adapter
- Name: VirtualBox Host-Only Ethernet Adapter

Shared Folders:

- Click the + icon (Add new shared folder)
- Folder Path C:\VMShare
- Folder Name VMShare
- tick Auto-mount

5. Click OK

10.3 - Install Windows Server 2022

1. Start the DC01-WindowsServer VM
2. Select English (United Kingdom) for the time format (this will automatically update the keyboard setting to United Kingdom), Next
3. Install Now
4. Select operating system: Windows Server 2022 Standard Evaluation (Desktop Experience)
5. Accept the licence, Next
6. Custom: Install Windows only
7. Select the unallocated drive, Next
8. Wait 15-25 minutes. When prompted, set a strong Administrator password

10.4 - Configure Network via Network Settings

1. Sign in as Administrator
2. Press Win + R, type ncpa.cpl, press Enter
3. Right-click Ethernet 2, Properties
4. Select Internet Protocol Version 4 (TCP/IPv4), Properties
5. Select Use the following IP address:
 - IP address: 192.168.56.12
 - Subnet mask: 255.255.255.0
 - Default gateway: leave blank
6. Preferred DNS server: 192.168.56.12 (the DC is its own DNS server)
7. Click OK, Close

10.5 - Update the Time Zone (if needed)

1. If the time is not correct, click on the search bar and search for 'Date & type settings'
2. Change the Time zone to match your region

10.6 - Rename the Server via System Properties

1. Right-click Start, System
2. Advanced system settings
3. Computer Name tab, Change
4. Under Computer name type DC01
5. Click OK, OK, Close, Restart Now

After restart, log back in as Administrator.

10.7 - Install Guest Additions and Map the Shared Folder

1. In the VirtualBox menu bar on the DC01-WindowsServer VM, select Devices, Insert Guest Additions CD Image
2. Open the **CD (D:)** drive in File Explorer inside the Server VM and run VBoxWindowsAdditions-amd64.exe, accept defaults, Reboot Now
3. After reboot, log in, then in the VirtualBox menu bar, select Devices, Shared Clipboard, Bidirectional. Also select Devices, Drag and Drop, Bidirectional. You can now copy and paste & drag and drop to/from this VM and your host PC
4. open PowerShell as Administrator and run:

```
net use V: \\vboxsvr\VMShare /persistent:yes
```

10.8 - Allow Communication between the VMs.

1. Open Powershell as Administrator and run:
`Get-NetConnectionProfile`
2. Ensure the NetworkCategory for Ethernet 2 is set to Private. If it is not, run this command:
`Set-NetConnectionProfile -InterfaceAlias "Ethernet 2" -NetworkCategory Private`
3. While still in Powershell, run:
`Get-NetFirewallRule -DisplayName "File and Printer Sharing (Echo Request - ICMPv4-In)" | Select Enabled, Profile, Direction, Action`
4. Ensure that it says 'True' underneath 'Enabled'. If not, run this command:
`Enable-NetFirewallRule -DisplayName "File and Printer Sharing (Echo Request - ICMPv4-In)"`
5. Verify that you can ping the Windows11-Endpoint VM (it must be powered on) from the DC01-WindowsServer VM:
`ping 192.168.56.11`
6. Verify that you can ping the DC01-WindowsServer VM (it must be powered on) from the Windows11-Endpoint VM:
`ping 192.168.56.12`

Take a VM snapshot: Clean Windows Server Install

11 - Active Directory Domain Services

11.1 - Install AD DS and DNS via Server Manager

1. In Server Manager click Manage, Add Roles and Features
2. Click Next through the first two screens (leave defaults)
3. Confirm the server is DC01, Next
4. Server Roles: tick Active Directory Domain Services
5. A popup asks to add required features, click Add Features
 - Also tick DNS Server, click Add Features
 - Click Next, Next, Next, Next, Install. Wait for install to complete
6. When complete click Close (do not tick automatic restart)

11.2 - Promote to Domain Controller via the Wizard

1. In Server Manager click the yellow notification flag at the top
2. Click Promote this server to a domain controller
3. Deployment Configuration: select Add a new forest, Root domain name: TestDomain.ie, Next
4. Domain Controller Options:
 - Leave functional levels at default
 - Leave DNS server ticked
 - DSRM Password: set a password and write it down separately
 - Next
5. DNS Options: Next (ignore the delegation warning)
6. Additional Options: wait for TESTDOMAIN to auto-populate as the NetBIOS name, Next
7. Paths: leave defaults, Next
8. Review Options: Next
9. Prerequisites Check: wait for it to complete, click Install
10. The server restarts automatically (3-5 minutes)

After restart, log in as TESTDOMAIN\Administrator.

11.3 - Verify Active Directory

Open PowerShell as Administrator and run:

```
Get-ADDomain  
nltest /dsgetdc:TestDomain.ie
```

You should see TestDomain.ie in the output and DC01 listed.

11.4 - Configure Account Lockout Policy via Group Policy

1. Press Win + R, type gpmmc.msc, press Enter
2. Expand Forest: TestDomain.ie, Domains, TestDomain.ie
3. Right-click Default Domain Policy, Edit
4. Navigate to: Computer Configuration, Policies, Windows Settings, Security Settings, Account Policies, Account Lockout Policy
5. Double-click Account lockout threshold, set to 5, click OK
6. Accept the suggested values for lockout duration and observation window (30 minutes each)
7. Close the Group Policy editor and GPMC

Take a VM snapshot: AD Domain Configured

12 - Create Domain Users and Organisational Units

12.1 - Create OU Structure via Active Directory Users and Computers

1. Press Win + R, type dsa.msc, press Enter
2. Expand TestDomain.ie in the left panel
3. Right-click TestDomain.ie, New, Organizational Unit
4. Name: SOC-Lab, OK
5. Right-click the new SOC-Lab OU, New, Organizational Unit and create each department inside it (one at a time): IT, HR, Finance, Marketing, Operations, Sales, Legal, R&D, Support, Management
6. Right-click TestDomain.ie, New, Organizational Unit
7. Name: Privileged-Accounts, OK

12.2 - Transfer lab-users.csv to the DC

1. On the Windows11-Endpoint VM, copy the lab.users.csv file from C:\Tools to the VMShare drive
2. on the DC01-WindowsServer VM, create a new folder in C: and name it "Tools"
3. copy lab.users.csv from the VMShare drive to C:\Tools

12.3 - Create Domain Users from the CSV

On the DC01-WindowsServer VM, Open PowerShell as Administrator and run:

```
$users = Import-Csv "C:\Tools\lab-users.csv"

$departments =
@("IT","HR","Finance","Marketing","Operations","Sales","Legal","R&D","Support","Management")

foreach ($user in $users) {
    if (Get-ADUser -Filter "SamAccountName -eq '$($user.Username)'" -ErrorAction SilentlyContinue) {
        Write-Host " [SKIP] $($user.Username)" -ForegroundColor Gray; continue
    }

    $dept    = $departments | Get-Random
    $ouPath  = "OU=$dept,OU=SOC-Lab,DC=TestDomain,DC=ie"
    $securePass = ConvertTo-SecureString $user.Password -AsPlainText -Force
    try {
        New-ADUser -Name $user.FullName -SamAccountName $user.Username `
            -UserPrincipalName "$($user.Username)@TestDomain.ie" `
            -AccountPassword $securePass -Enabled $true -Path $ouPath `
            -Department $dept -Description "SOC Lab - $dept" `
            -PasswordNeverExpires $true -ErrorAction Stop

        Write-Host " [+] TESTDOMAIN\$($user.Username)" -ForegroundColor Green
    } catch { Write-Host " [!] Failed: $($user.Username) - $_" -ForegroundColor Red }
}

$adminUsers = $users | Where-Object { $_.IsAdmin -eq "Yes" }

foreach ($admin in $adminUsers) {
    Add-ADGroupMember -Identity "Domain Admins" -Members $admin.Username -ErrorAction
    SilentlyContinue

    Write-Host " [*] Domain Admin: TESTDOMAIN\$($admin.Username)" -ForegroundColor Yellow
}

Write-Host "Done." -ForegroundColor Cyan
```

13 - Join Windows 11 to the Domain

13.1 - Configure DNS to Point to the Domain Controller

Windows 11 must use the DC as DNS on both adapters. TestDomain.ie is a real domain on the public internet so without pointing to your DC, it resolves to the wrong address.

For Ethernet (NAT adapter):

1. Right-click Ethernet, Properties, IPv4, Properties
2. Preferred DNS server: 192.168.56.12
3. Click OK, Close

For Ethernet 2 (Host-Only adapter):

1. Press Win + R, type ncpa.cpl, press Enter
2. Right-click Ethernet 2, Properties
3. Select IPv4, Properties
4. Preferred DNS server: 192.168.56.12
5. Click OK
6. Untick Internet Protocol Version 6 (TCP/IPv6)
7. Close

Run in PowerShell:

`Clear-DnsClientCache`

`Resolve-DnsName TestDomain.ie`

The result must include 192.168.56.12. If it still shows a public IP, recheck both adapters.

13.2 - Join the Domain via System Properties

1. Right-click Start, System
2. Advanced system settings
3. Computer Name tab, Change
4. Under Member of, select Domain
5. Type: TestDomain.ie
6. Click OK
7. Enter credentials:
 - Username: TESTDOMAIN\Administrator
 - Password: **your DC Administrator password**
8. Click OK
9. After a few seconds, you should see a "Welcome to the TestDomain.ie domain." message
10. Click OK, OK, Close, Restart Now

If the join fails with request is not supported, run this in PowerShell on DC01-WindowsServer, then try joining the domain again:

`Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False`

13.3 - Verify Domain Membership

After restart, log in as TESTDOMAIN\Administrator. Open PowerShell as Administrator and run:

```
(Get-WmiObject Win32_ComputerSystem).Domain
```

This should return TestDomain.ie

```
klist
```

This should show Kerberos tickets for krbtgt/TESTDOMAIN.IE.

13.4 - Restore NAT Adapter DNS

1. Press Win + R, type ncpa.cpl
2. Right-click Ethernet (NAT), Properties, IPv4, Properties
3. Set Preferred DNS server to 8.8.8.8
4. Click OK, Close

13.5 - Install RSAT Active Directory Tools

1. Start, Settings System, Optional Features
2. View features, See available features
3. Search for RSAT
4. Click RSAT: Active Directory Domain Services and Lightweight Directory Services Tools
5. Add
6. Wait 1-2 minutes for installation

Take snapshots of both VMs: Windows 11 Joined to TestDomain.ie

14 - Splunk Universal Forwarder on DC01

14.1 - Install the Splunk UF onto the DC

1. On DC01 VM, copy the splunkforwarder MSI from the VMShare drive to the C:\Tools folder
2. Double-click the MSI in C:\Tools and follow the same screens as Windows 11:

Screen 1 - Licence:

- Tick Check this box to accept the License Agreement
- Leave An on-premises Splunk Enterprise instance selected
- Click Next

Screen 2 - Credentials:

- Username: admin
- Password: choose something, note it down (this is just for the forwarder's own management interface)
- Click Next

Screen 3 - Deployment Server:

- Leave completely blank
- Click Next

Screen 4 - Receiving Indexer:

- Hostname or IP: 192.168.56.10
- Port: 9997
- Click Add then Next

Screen 5 - Install, Finish

14.2 - Configure inputs.conf via Notepad

1. In File Explorer, go to **C:\Program Files\SplunkUniversalForwarder\etc\system\local**
2. Ensure that File name extensions are enabled in the View menu
3. Create a new Text document and name it: inputs.conf
4. Right-click the new inputs.conf file, Open with, Notepad, paste in:

[WinEventLog://Security]

index = dc-logs

disabled = 0

renderXml = false

[WinEventLog://System]

index = dc-logs

disabled = 0

renderXml = false

[WinEventLog://Directory Service]

index = dc-logs

disabled = 0

renderXml = false

[WinEventLog://DNS Server]

index = dc-logs

disabled = 0

renderXml = false

[WinEventLog://DFS Replication]

index = dc-logs

disabled = 0

renderXml = false

5. Save it.

14.3 - Configure outputs.conf via Notepad

1. Open the outputs.conf file (from the same location) with Notepad
2. Replace all the existing text with this:

[tcpout]

defaultGroup = splunk_indexer

[tcpout:splunk_indexer]

server = 192.168.56.10:9997

compressed = true

[tcpout-server://192.168.56.10:9997]

3. Save it.

14.4 - Run Forwarder as Local System via Services

1. Press Win + R, type services.msc, press Enter
2. Find SplunkForwarder
3. Right-click, Properties, Log On tab
4. Select Local System account if it is not already selected, click OK
5. Right-click SplunkForwarder, Stop If error 1053 appears, open PowerShell as Administrator and run: taskkill /F /IM splunkd.exe
6. Right-click SplunkForwarder, Start

14.5 - Verify DC Logs are Flowing

In Splunk Web (On your host PC), search:

```
index=dc-logs earliest=-1hr  
| stats count by source
```

You should see WinEventLog:Security, WinEventLog:System, and WinEventLog:Directory Service.

The Directory Service log is very low volume. If you don't see any WinEventLog:Directory Service events, run this command in powershell on the DC01-WindowsServer VM:

```
Get-WinEvent -LogName "Directory Service" -MaxEvents 5
```

Check the time for the events shown and expand the search to suit. So for example, if the latest events shown were 3 hours ago, change the search to this:

In Splunk Web, search:

```
index=dc-logs earliest=-3hr  
| stats count by source
```

15 - Domain Activity Simulator

15.1 - Copy the Domain Simulator to Windows 11

1. Copy domain-simulator.ps1 (find it with the github link given at the start of this guide) to **C:\VMShare** on your host PC.
2. Inside the Windows 11 VM, copy and paste it from the **VMShare** drive to **C:\Tools**

15.2 - Test the Script

In PowerShell, run:

```
powershell.exe -ExecutionPolicy Bypass -File "C:\Tools\domain-simulator.ps1"
```

You should see domain login scenarios appearing. Press Ctrl+C to stop.

15.3 - Register the Domain Simulator Scheduled Task

1. Register the scheduled task pointing at domain-simulator.ps1

```
$action = New-ScheduledTaskAction -Execute "powershell.exe" -Argument "-WindowStyle  
Hidden -ExecutionPolicy Bypass -File C:\Tools\domain-simulator.ps1" $trigger = New-  
ScheduledTaskTrigger -AtStartup $settings = New-ScheduledTaskSettingsSet -  
ExecutionTimeLimit 0 -RestartCount 3 -RestartInterval (New-TimeSpan -Minutes 1) $principal =  
New-ScheduledTaskPrincipal -UserId "SYSTEM" -LogonType ServiceAccount -RunLevel Highest  
Register-ScheduledTask -TaskName "SOC-Lab-Activity-Simulator" ` -Action $action -Trigger  
$trigger -Settings $settings -Principal $principal Write-Host "Domain simulator registered." -  
ForegroundColor Green
```

2. Kick it off immediately rather than waiting for next reboot

```
Start-ScheduledTask -TaskName "SOC-Lab-Activity-Simulator"
```

3. Confirm it's running correctly

```
Start-Sleep -Seconds 5 Get-CimInstance Win32_Process -Filter "Name = 'powershell.exe'" |  
Select-Object ProcessId, CommandLine Get-ScheduledTaskInfo -TaskName "SOC-Lab-Activity-  
Simulator"
```

The Domain Simulator script will automatically run as long as the VM is powered on. All activity will be logged in **C:\Tools\domain-simulator-log.txt**. Use this log file to confirm your investigations into domain activity in Splunk Web.

16 – Splunk Dashboards and Alerts

16.1 - Important Note About Domain Login Events

Failed domain login attempts generate Event ID 4625 in windows-logs (on Windows 11) rather than Event ID 4771 in dc-logs (on the DC). This is because Windows 11 records interactive logon failures locally via NTLM. This is normal in many real enterprise environments.

The DC logs (dc-logs) are valuable for successful Kerberos events (4768), account lockouts (4740), and AD changes (4720, 4728, 4729).

16.2 - Create Dashboard

1. In Splunk web, go to Search & Reporting, Dashboards, Create New Dashboard
2. For the title, enter "SOC Overview", leave the rest as default and select 'Classic Dashboards', then 'Create'. We will now add 11 panels which will give you a good starting point to analyse domain activity from.
3. For each of the below panels, start by clicking the 'Add panel' button, then open the 'New' drop down menu, select the panel type (in parenthesis), enter in the content title (the name of the panel, in bold) and search string (in red), then click 'Add to Dashboard'

Failed Logins Over Time (Line Chart)

(index=windows-logs OR index=dc-logs) EventCode=4625 | timechart count as "Failed Logins" span=5m by index

Top Failed Login Accounts (Bar Chart)

(index=windows-logs OR index=dc-logs) EventCode=4625 | stats count as "Failed Attempts" by Account_Name | sort -"Failed Attempts" | head 10

Failed Logins details (Statistics Table)

(index=windows-logs OR index=dc-logs) EventCode=4625 | stats count by Account_Name, Failure_Reason, host, index | sort -count | rename Account_Name as "Username", count as "Failed Attempts"

Brute Force Detected (Statistics Table)

(index=windows-logs OR index=dc-logs) EventCode=4625 earliest=-24h | bin _time span=10m | stats count by _time, Account_Name, index | where count >= 5 | eval "Time Window"=strftime(_time,"%Y-%m-%d %H:%M") | table "Time Window", Account_Name, count, index | rename Account_Name as "Target Account", count as "Failed Attempts" | sort -"Failed Attempts"

Domain Account Lockouts (Statistics Table)

index=dc-logs EventCode=4740 earliest=-24h | table _time, Account_Name, Caller_Computer_Name, host | sort -_time

Successful Kerberos Authentication (Statistics Table)

```
index=dc-logs EventCode=4768 Result_Code="0x0" earliest=-24h | rex field=Client_Address  
"::ffff:(?<Client_Address>\d+\.\d+\.\d+\.\d+)" | table _time, Account_Name, Client_Address, host |  
sort -_time
```

New User Created (Statistics Table)

```
(index=windows-logs OR index=dc-logs) EventCode=4720 | table _time, index, Account_Name,  
Subject_Account_Name, host | sort -_time
```

Domain Admin Group Changes (Statistics Table)

```
index=dc-logs (EventCode=4728 OR EventCode=4729) earliest=-24h | eval  
Action=if(EventCode=4729,"REMOVED from group","ADDED to group") | table _time, Action,  
Member_Name, Group_Name, Subject_Account_Name, host | sort -_time
```

Recent Process Creations (Statistics Table)

```
index=sysmon-logs EventCode=1 | table _time, User, Image, CommandLine, ParentImage, host |  
sort -_time
```

Outbound Connections (Statistics Table)

```
index=sysmon-logs EventCode=3 | table _time, User, Image, DestinationIp, DestinationPort, host |  
sort -_time
```

Suspicious PowerShell (Statistics Table)

```
index=windows-logs EventCode=4104 | search Message="EncodedCommand" OR  
Message="Invoke-Expression" OR Message="DownloadString" OR Message="FromBase64String"  
| search NOT Message="Domain Activity Simulator" | table _time, host, Message | sort -_time
```

4. Once all panels have been added, close the 'Add panel' menu and then click 'Save'

16.3 - Create Alerts

1. Go to Search & Reporting, Search
2. Search each of the below search strings (in red). After each search, click Save as, Alert. Enter the Title (in bold), change 'Run every week' to 'Run on Cron Schedule'. Select "Last 15 minutes" for the Time Range. For the Cron Expression, enter: *** / 5 * * * ***. Leave everything else as default. Under 'Trigger Actions', click 'Add Actions', Add to Triggered Alerts, select the severity (in parenthesis). Click 'Add Actions' again, Run a script, enter **slack_alert.sh** for the file name. click 'Save'

Brute Force (High)

```
(index=windows-logs OR index=dc-logs) EventCode=4625 | bin _time span=10m | stats count by  
_time, Account_Name, index | where count >= 5
```

Domain Account Lockout (Medium)

`index=dc-logs EventCode=4740`

Domain Admin Group Changes (Critical)

`index=dc-logs (EventCode=4728 OR EventCode=4729)`

New Domain User Created (Medium)

`index=dc-logs EventCode=4720`

PowerShell Encoded Command (High)

`index=windows-logs EventCode=4104 | search Message="EncodedCommand"`

AD Recon via PowerShell (High)

`index=windows-logs EventCode=4104 | search Message="Get-ADUser" OR Message="Get-ADGroup" OR Message="Get-ADComputer" OR Message="Get-ADDomain"`

17 - Slack Workplace for Splunk Alerts

17.1 - Create Channel

1. Go to slack.com/get-started and enter in your email, verify with the code you receive
2. Name the workspace SOC Lab
3. Enter in a Username (whatever you like)
4. Skip inviting teammates
5. Create a channel called #soc-lab-alerts

17.2 - Create Webhook

1. Go to api.slack.com/apps and select Create an App > Blank app
2. Name it Splunk SOC Lab and select your SOC Lab workspace, click Create
3. In the app's settings sidebar, click Incoming Webhooks (under 'Features')
4. Toggle Activate Incoming Webhooks to On
5. Scroll down, click Add New Webhook
6. Select the #soc-lab-alerts channel, Allow
7. You'll now see a Webhook URL — copy it to a notepad (you'll need it later)

17.3: Create the slack_alert.sh script

1. On the Ubuntu-SplunkServer VM, run each of the below commands in order:

```
sudo apt install -y jq
```

```
sudo mkdir -p /opt/splunk/bin/scripts
```

```
sudo nano /opt/splunk/bin/scripts/slack_alert.sh
```

2. Paste the below text into the nano editor (making sure to add in your webhook URL from the previous step):

```
#!/bin/bash
WEBHOOK_URL="YOUR WEBHOOK URL"

EVENT_COUNT="{1:-Unknown}"
ALERT_NAME="{4:-Unknown Alert}"
RESULTS_LINK="{6:-N/A}"
TIMESTAMP="$(date -u +%Y-%m-%d %H:%M:%S UTC)"

case "$ALERT_NAME" in
    "Brute Force") SEVERITY="HIGH" ;;
    "Domain Account Lockout") SEVERITY="MEDIUM" ;;
    "Domain Admin Group Changes") SEVERITY="CRITICAL" ;;
    "New Domain User Created") SEVERITY="MEDIUM" ;;
    "PowerShell Encoded Command") SEVERITY="HIGH" ;;
    "AD Recon via PowerShell") SEVERITY="HIGH" ;;
    *) SEVERITY="UNKNOWN" ;;
esac

case "$SEVERITY" in
    "CRITICAL") COLOR="#8B0000"; EMOJI=":rotating_light:" ;;
    "HIGH") COLOR="#FF0000"; EMOJI=":warning:" ;;
    "MEDIUM") COLOR="#FFA500"; EMOJI=":large_orange_circle:" ;;
    *) COLOR="#808080"; EMOJI=":grey_question:" ;;
esac

MESSAGE_TEXT="{EMOJI} *[{SEVERITY}] {ALERT_NAME}*
*Event Count:* {EVENT_COUNT}
*Time:* {TIMESTAMP}
*View Results:* {RESULTS_LINK}"

PAYLOAD=$(jq -n \
--arg color "$COLOR" \
--arg text "$MESSAGE_TEXT" \
'{attachments: [{color: $color, text: $text}]}')

curl --cacert /etc/ssl/certs/ca-certificates.crt -s -X POST -H 'Content-type:
application/json' --data "$PAYLOAD" "$WEBHOOK_URL" > /dev/null
```

3. Press Ctrl+X, then Y, then enter

4. Enter the below commands in order:

```
sudo chmod +x /opt/splunk/bin/scripts/slack_alert.sh
```

```
sudo chown UbuntuAdmin:UbuntuAdmin /opt/splunk/bin/scripts/slack_alert.sh
```

```
/opt/splunk/bin/scripts/slack_alert.sh 5 "" "" "Brute Force" "" http://test-link
```

5. Ensure that a properly formatted message appears in your #soc-lab-alerts channel
6. On your host PC, open Notepad as Administrator. File, Open
`C:\Windows\System32\drivers\etc\hosts` (you will need to change the file type to All files to be able to see the hosts file)
Paste in this at the bottom of the hosts file:
192.168.56.10 ubuntu-splunkserver

You should now have an operational Slacks channel where triggered alerts will appear, colour coded based on their severity

Conclusion

With all steps complete, the SOC Analyst lab is now fully operational and ready for use. We now have three interconnected VMs that work off each other to create Windows event logs and Sysmon data that can be seen and analysed in Splunk through a custom-built dashboard and alert system that uses a slack channel to notify you of triggered alerts that you can investigate. The lab is quite basic by design, as it is only meant to serve as a starting point for those looking to get their first hands-on experience with a SOC environment. However, it also serves as a solid foundation to build a more advanced lab on top of.